Pranay Grandhi

pranaygrandhi9@gmail.com | +91-8297380938 | github.com/pranaygrandhi | linkedin.com/in/pranay-grandhi

PROFESSIONAL SUMMARY

Computer Science and Cyber Security undergraduate with a strong foundation in cybersecurity, network security, vulnerability assessment, web application security, authentication and authorization, encryption fundamentals, security monitoring, incident response concepts, and secure system design. Hands-on experience in vulnerability assessment, penetration testing, network traffic analysis, attack identification, and security analysis using Wireshark, Nmap, Burp Suite, and Linux. Knowledge of TCP/IP, DNS, HTTP/HTTPS, firewalls, common network threats, OWASP Top 10, security controls, and security best practices.Demonstrated analytical, problem-solving, research, documentation, and communication skills by identifying and addressing vulnerabilities in application, system, and network environments, reflecting a strong commitment to security best practices.

EDUCATION

SRM Institute of Science and Technology 2024 – Present
B.Tech – Computer Science and Engineering (Cyber Security) *CGPA: 9.33*

Intermediate (Class XII) **83%**

Secondary School (Class X) **93.1%**

EXPERIENCE

Network Security Associate Intern – Fortinet Jul 2024 – Oct 2024
EduSkills

- Analyzed network traffic and identified potential security vulnerabilities using industry-standard security tools.
- Applied firewall, IDS, network security, TCP/IP, and basic network security concepts to strengthen security analysis capabilities.
- Studied network configurations, security controls, common network threats, and attack patterns in security-focused environments.

Ethical Hacking Intern Jul 2025 – Oct 2025
EduSkills

- Conducted vulnerability assessments and penetration testing in simulated and authorized environments to identify security weaknesses.
- Applied ethical hacking methodologies to analyze vulnerabilities and evaluate application and system security.
- Identified security weaknesses and proposed mitigation and remediation strategies following security best practices.
- Applied ethical hacking methodologies, including penetration testing and code analysis, to identify vulnerabilities and strengthen application and system security, resulting in enhanced overall protection.

CYBERSECURITY PROJECTS

Security Analysis of Internet-Based Voice and Video Communication Infrastructure

- Analyzed SIP and RTP VoIP protocols across 100+ packets using Wireshark for network and security analysis.
- Performed packet-level traffic analysis to identify unusual communication patterns and security weaknesses.
- Identified Man-in-the-Middle (MITM) and replay attack scenarios through protocol and network traffic analysis.
- Implemented Secure RTP (SRTP) and TLS encryption to improve communication security.
- Evaluated secure communication mechanisms and encryption fundamentals for protecting network traffic.

Practical Attacks and Countermeasures for Cryptosystems

- Simulated brute-force, timing, and cryptographic attacks to analyze vulnerabilities in cryptographic systems.
- Designed security countermeasures using robust key management and encryption techniques.
- Evaluated attack scenarios and recommended security improvements to reduce vulnerability exposure.

Security Analysis of CPU–GPU Coupled Architectures

- Investigated security vulnerabilities in heterogeneous CPU–GPU architectures.
- Identified memory leakage and side-channel attack risks affecting system security.
- Proposed secure memory isolation techniques to improve system resilience and reduce security exposure.

ADDITIONAL PROJECT EXPERIENCE

IN-SPACe Rocketry Student Competition – Critical Design Review Qualifier

- Successfully advanced to the Critical Design Review (CDR) by passing the Preliminary Design Review (PDR) in the national-level IN-SPACe Rocketry Student Competition.
- Collaborated with a multidisciplinary engineering team on the design and development of a student-built sounding rocket.
- Contributed to subsystem design, system integration, engineering calculations, and technical documentation.
- Participated in design validation, risk assessment, trade-off analysis, and preparation of technical reports and presentations.

TECHNICAL SKILLS

Cybersecurity: Vulnerability Assessment, Penetration Testing, Ethical Hacking, Web Application Security, Network Security, System Security, Threat Analysis, Security Controls, Security Monitoring, Incident Response Concepts, Information Security

Network Security: TCP/IP, DNS, HTTP/HTTPS, Firewalls, IDS, Network Traffic Analysis, Common Network Threats, Attack Patterns, Secure Communication

Application Security: Authentication, Authorization, Input Validation, Secure Coding Fundamentals, OWASP Top 10, Web Application Security

Security Tools and Platforms: Linux, Wireshark, Nmap, Burp Suite, Git, GitHub, Visual Studio Code

Security Concepts: Encryption Fundamentals, Cryptography, Vulnerability Management, Security Best Practices, Risk Assessment, Security Analysis

Programming Languages: Python, C, C++, Java

SOFT SKILLS

Analytical Thinking , Problem Solving , Technical Documentation , Communication , Continuous Learning , Critical Thinking , Time Management, Curiosity

CERTIFICATIONS AND TRAINING

Cybersecurity Professional , Cybersecurity Associate - SOC Analyst (Ongoing) , Fortinet Certified Associate (FCA) , Fortinet Certified Fundamentals (FCF) , Cisco Network Defence , ISO/IEC 27001:2022 Lead Auditor